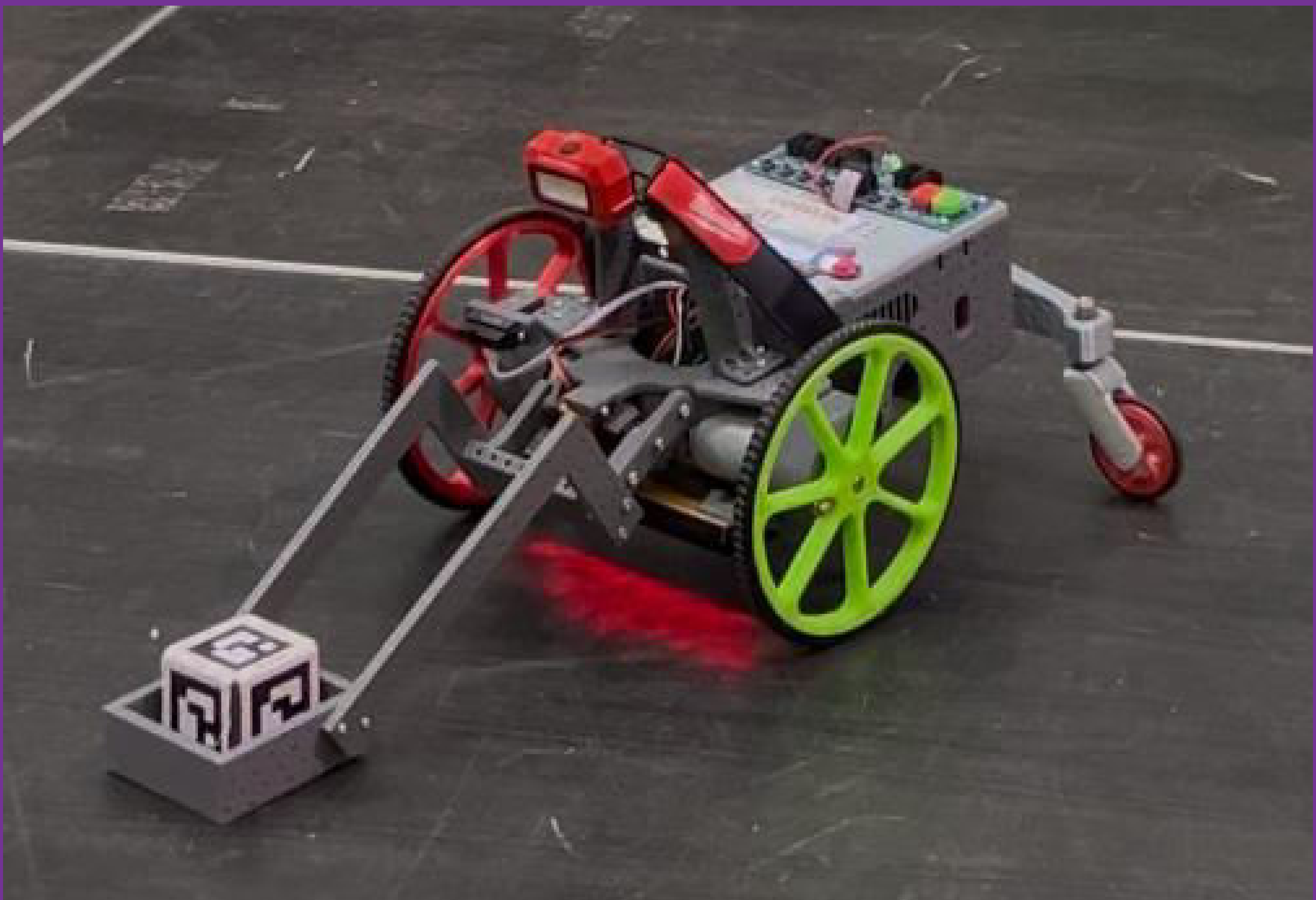
DTU ELECTRO
Department of Electrical and Photonics Engineering

Challenge report
Group 119 Newton

34755 Building dependable robot systems

Lukas Egkher

2025



Challenge report
Group 119 Newton

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Course	34755 Building dependable robot systems
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Contents

Chapter 1

Introduction

1.1 The group

The group consisted of three members.

1. s243430 Lukas Egkher (me, CTO).
2. s242705 Jóel Ingason (CPO, CLO).
3. 242739 Teitur Tannason (CCO).

1.2 DTU Robocup performance

The result of the competition was 14 points in the first run in the finals and 1 point in the second run. The first run was satisfactory only to a certain point, since we knew that the three-gate was not 100% reliable, but we still decided to risk it and it failed. The second run then was a huge disappointment, as it seemed that the battery had some issues (which happened a lot during the testing phase). The robot crashed after passing through the first gate.

We had prepared other challenges, including the mini-golf and going down the stairs, but the stairs were too unsafe and the mini-golf not tested at a certain level, and therefore both were omitted.

1.3 Planning tools

To develop our RoboCup robot, we initially decided to use the Scrum methodology. The idea was to remain flexible and to make steady progress through short and focused development phases. Although we did not follow Scrum to the letter, the general approach of working in iterations and adjusting our plans as we went proved helpful.

Our plan was to hold regular weekly meetings to discuss progress and align on the next steps. In practice, it was not always easy to get everyone together due to scheduling conflicts and other commitments. As a result, much of our communication ended up happening over WhatsApp, where we kept each other updated and coordinated informally. Everyone mainly focused on their own tasks, but we stayed in touch enough to make sure that all the pieces came together. Especially for functionality used over different tasks and branches it was important to inform team members when changes have been pushed on the main branch, so everyone is up-to-date.

Looking back, our mix of structured planning and flexible and independent work helped us make the most of our time and resources. It wasn't always perfectly organized, but the approach worked for us and allowed us to continue moving forward, even when things became stressful.

1.4 Team performance

As mentioned above, we barely worked together the first half of the course, which was still fine for us. The more time passed, the more we understood the benefit of working on tasks together. This improved finding a suitable solution faster by using everything we have implemented already.

Especially during the Easter break we spend days and nights together in the library working a lot, which also improving the performance of our robot more than before.

In terms of effort distribution, we tried to balance the workload, and while each of us had different strengths and commitments, we all put in significant time - particularly in the final phase.

Overall, the team worked very well, and we grew a lot in terms of working in a group combining a lot of knowledge.

1.5 Lessons learned

One key lesson learned was the value of consistent communication and collaboration. While we were initially more independent in our work, the later stage of the project showed us how productive group cooperation can be. Spending focused time together improved both our technical results and group morale. We also learned how important it is to plan for uncertainties - such as hardware issues or scheduling constraints - and adapt quickly.

1.6 The robot

We used our robot 119 Newton for the whole course. We had issues with one of the line sensors once but Jóel was able to fix it again. Our gripper for the golf-ball and luggage delivery task had three different designs and in the picture the final version can be seen.

1.7 My challenge

My role was do all the hardware design. Furthermore, I contributed to most challenges, but primarily focused on the area around the sorting center, including the luggage delivery task and the axe gate. I was present at the competition on qualification day but missed the finals due to my work. My individual challenge is the delivery of luggage. This task was executed only in the finals because it was not 100% reliable and there was no need to take a huge risk on the qualification day.

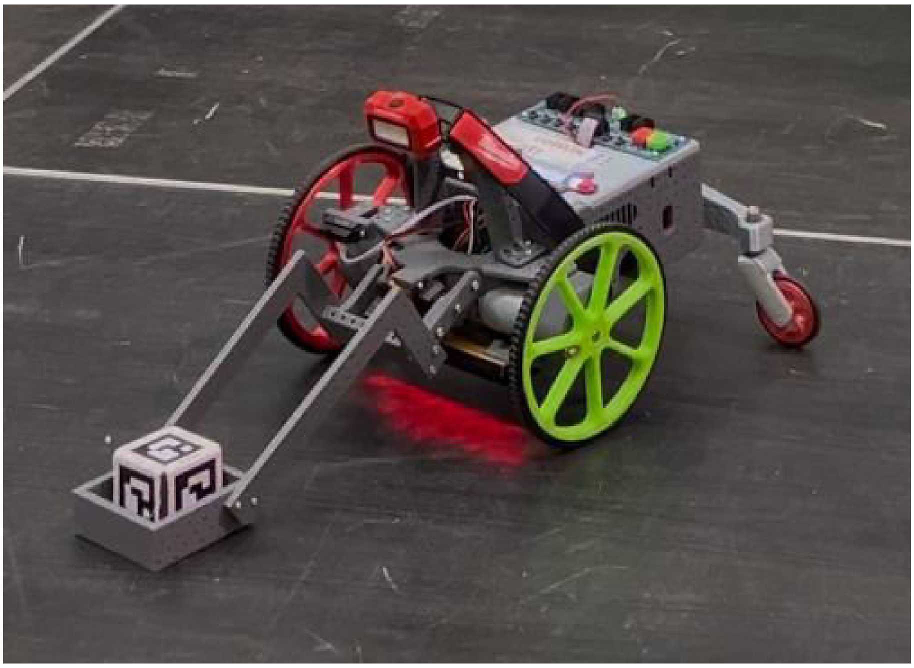


Figure 1.1: The robot used for all testing equipped with the final version of the gripper during the luggage delivery task.

Chapter 2

The challenge

2.1 Introduction

The goal is to navigate from finishing the stairs task to the yellow shuttle, position the robot accordingly, and bring the luggage cubes down. Afterwards, it is necessary to detect the correct cube based on the ArUco code and navigate the robot with the grabbed cube towards the 'D'-field in the sorting center.

The main uncertainty is getting the luggage cubes off in a reliable way, so it is possible to detect and pick them up.

2.2 Challenge analysis

The challenge will start after coming from the stairs and end in readiness for the axe gate challenge.

There are a maximum of 4 points on the luggage delivery if the correct luggage is brought to the correct place, otherwise less points are given. All 4 points should be achieved.

We kept our initial idea as simple as possible. The first big issue was to push the luggage cubes down on the correct side of the wall. Therefore, the robot must be close enough to reach them with the gripper. Simple trial and error showed that the front edges of the gripper should point slightly to the outside. This increased the success rate of getting the luggage cubes to the correct side by a lot.

The second issue was to determine the direction the robot should go after picking up the luggage cube. By turning constantly until an ArUco code from the sorting center is detected, it is possible to navigate the robot safely to field 'D'.

Several alternative approaches were considered. One idea was to track the yellow shuttle using computer vision and approach it dynamically, but we opted against it due to complexity and risk of inconsistency. We chose the simplest reliable method: fixed positioning and a timed push. This reduced randomness and made the task more repeatable under competition conditions.

2.3 The solution

The challenge starts at A in figure 2.1, and the heading is oriented towards the white number 3 and the red extra time sensor. The robot follows the line until its end and from there drives 30cm forward and turns 70 degrees towards the right.

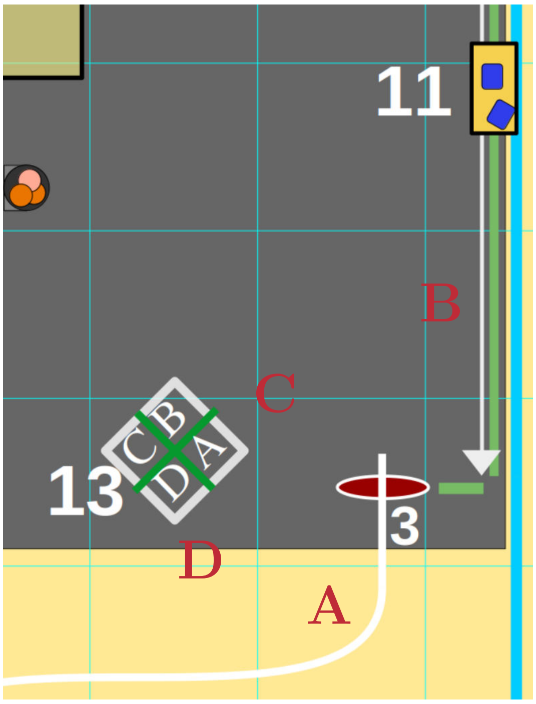


Figure 2.1: The luggage delivery is approached from point A by following the white line, then moving towards position B using first fixed values and then the IR sensor to get an ideal position. After picking up the luggage cube the robot moves towards C using the ArUco codes on the sorting center to orientate itself, and finally finding its way to point D where the next challenge starts after dropping off the cube.

At position B the robot uses the IR sensor to get the ideal distance to the green wall and is waiting for the shuttle to arrive as in Figure 2.2a.

Once the ArUco code is detected at a certain position in the camera frame, the gripper is set low and the robot does a left turn immediately in order to push the correct cube off the yellow shuttle.

The robot then reverses slightly and tries to relocate the ArUco code, allowing it to properly grab the cube on the second attempt, as it can be seen in Figure 1.1.

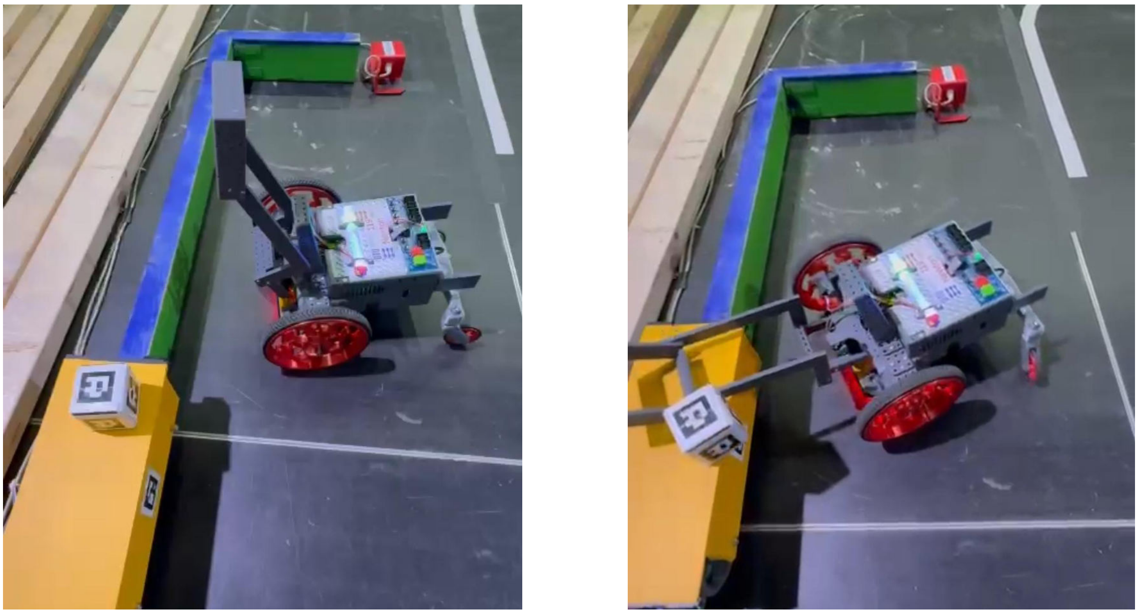
Furthermore, the robot turns around until the 'A'-field in the sorting center (white number 13 in Figure 2.1) is detected through its ArUco code. The robot then approaches this code up to a certain pixel size and then moves from position C to position D by doing a hard coded mix of 90 degree turns and driving forward.

At position D the robot moves again by detecting the ArUco code until a certain distance is reached and then leaves the luggage cube there by simply lifting the gripper.

2.4 Performance analysis

In terms of reliability we had some issues with the distance driven towards position B in Figure 2.1 because sometimes the luggage cube falls so awful, that the yellow shuttle crushed into it, on its way back. Therefore, we increased this distance from 30 to 80 centimeters in order to get a little more time to pick up the cube before the shuttle comes back. Furthermore, we approach the shuttle closer to its center of movement range.

A big disadvantage of not tracking the shuttle but having a fixed position to approach



(a) Waiting position

(b) Turning position

Figure 2.2: Pushing the luggage cube of the shuttle with a turn.

it instead, is the enourmously long waiting time, when one just missed the shuttle by a margin. Approaching the shuttle more in the middle does not change the average time until the shuttle passes in front of the robot, but it decreases the maximum duration one may have to wait. And this is an advantage, as well, since we only want to pass the axe gate afterwards and then move towards the three-gate where more time is added again. Meaning, some waiting time is fine, if it is not something like 30 seconds.

A test has been executed 10 times to evaluate if time improvements are needed for the whole luggage delivery task. The time started the moment the robot passed the extra time sensor (red oval with number 3 in Figure 2.1).

Table 2.1 then shows the time in seconds it took to throw off the cube from the shuttle and the time to actually bring it to the sorting center. Failed attempts are marked with '-'. The results demonstrate both time consistency and success rates. Performance was tested under real conditions, with Figure 2.1 showing spatial references for each key step.

The final issue identified was with ensuring the luggage cube was securely inside the gripper. In a few cases the cube was not aligned in a suitable manner and the robot drove towards the sorting center without it. Therefore, a little wiggle was introduced once the gripper was send flat in order to make sure that the cube is actually inside the square of the gripper.

Overall, the luggage delivery challenge was performed reliably with a high success rate and within acceptable time margins. While a few execution risks remained - especially in cube alignment and shuttle timing - the approach proved suitable for the goals of our competition run.

2.5 Dependability analysis

The chosen solution showed high dependability, with a success rate around 90% and consistent performance under test conditions. The main hardware risk was the possibility of the cube being pushed off the shuttle in an unpredictable orientation, which could

Try	Cube Thrown Off (in sec)	Cube in Sorting Center (in sec)
1	14	30
2	21	39
3	17	34
4	26	-
5	20	37
6	13	30
7	24	41
8	27	45
9	19	37
10	22	40

Table 2.1: Observed times for luggage delivery task - time started when extra time sensor gets passed. '-' indicating that task failed

prevent proper gripping. However, the gripper design proved stable and reliable in most trials. The IR sensor and ArUco detection also worked consistently, though slight lighting changes could affect detection timing especially at night in the library.

Compared to alternative solutions - such as dynamically tracking the shuttle - our fixed-position strategy significantly reduced potential failure points. Real-time tracking would have introduced sensor complexity and timing uncertainty, which we actively avoided. Thus, while our solution was slightly less time-efficient, it offered greater stability and predictability.

2.6 Possible improvements

A recovery strategy could be implemented by actively checking whether the cube is secured after the grab attempt. If the robot fails to detect the cube (e.g., via a simple camera check), it could retry the approach. After three failed attempts or a time threshold, the robot should abandon the task and proceed to the axe gate. This would reduce the chance of the robot wasting valuable time in an unrecoverable state.

If the goal would have been to do the multi-ball task or the sentry as well, more time would have been necessary. Therefore, it might have been necessary to actually detect the shuttle, drive up to a short way in front of it and do the whole process at that specific position. That would have generated way more randomness since the overall distance towards the sorting center would have differed a lot.

More room for time improvement would have been in the way the robot approaches the 'D'-field in the sorting center. It is kinda jagged at the moment so a much smoother approach could have been implemented - it was just not necessary for the tasks we did.

2.7 Conclusion

The individual challenge was a kinda tricky since there seemed to be a lot of possible ways getting the luggage's down from the shuttle. I probably spent to much time thinking about the 'perfect' solution to it, since in the end a simple approach succeeded.

To conclude, more time for testing and also trying things out before overthinking to much would have benefited a lot.